

- Functional Requirements Document (FRD) for Data Platform**
- Introduction:**

The Data Platform is designed to integrate and aggregate customer and order data stored in CSV files. This document outlines the functional requirements for the Data Platform.

- Functional Requirements:**

1. **Data Ingestion**

- Read source CSV data from the specified volumes and load it into the delta tables "customer" and "order".
- The data ingestion process should handle CSV files located at:
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`.

2. **Data Schema Management**

- Create and manage the schema of the "customer" and "order" delta tables with the specified columns.
- The schema of the "customer" table is: `CustId`, `Name`, `EmailId`, `Region`.
- The schema of the "order" table is: `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId`.

3. **Data Cleansing**

- Remove null and duplicate records from both "customer" and "order" tables.

4. **Order Summary Table Creation and Management**

- Create the "ordersummary" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
- The schema of the "ordersummary" table is: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`.
- Implement SCD type 2 logic to update the "ordersummary" table whenever there is a change in the "customer" table.

5. **Customer Aggregate Spend Table Creation and Management**

- Create the "customeraggregatespend" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
- The schema of the "customeraggregatespend" table is: `Name`, `TotalAmount`, `Date`.
- Aggregate the "TotalAmount" column from the "ordersummary" table, grouping by the "Name" and "Date" columns, and load the aggregated data into the "customeraggregatespend" table.
- Data Flow Diagram:**

``json

```
{"name": "generate_dfd_diagram", "arguments": {"text": "CSV data is read from volumes and loaded into delta tables 'customer' and 'order'. The data is then cleansed and joined to create 'ordersummary' table with SCD type 2 logic. The 'ordersummary' table is then aggregated to create 'customeraggregatespend' table."}}
```

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- System Components:**
- Data Ingestion Component: responsible for reading CSV data and loading it into delta tables.
- Data Cleansing Component: responsible for removing null and duplicate records from delta tables.
- Order Summary Component: responsible for creating and managing the "ordersummary" table with SCD type 2 logic.
- Customer Aggregate Spend Component: responsible for creating and managing the "customeraggregatespend" table.
- Data Quality and Validation:**

The Data Platform should ensure data accuracy, completeness, and consistency throughout the data integration and aggregation process. Data quality checks should be performed after data cleansing and aggregation to verify the accuracy of the data.

- Assumptions and Dependencies:**

The input CSV files are in the correct format and location. The delta tables "customer" and "order" are created with the correct schema. The catalog "gen_ai_poc_databricksco" and schema "sdic_wizard" exist in the target data storage system.

- Success Criteria:**

The data is successfully loaded into the delta tables "customer" and "order". The "ordersummary" table is created and populated with the joined data. The SCD type 2 logic is implemented correctly, and the "ordersummary" table is updated accordingly. The "customeraggregatespend" table is created and populated with the aggregated data.